

Leonardo De Andrade Santos

Quantum Control / Automation Candidate | Python • Measurement Automation • RF/DSP • Embedded Systems

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Professional Summary

Electrical and embedded systems engineer with 4+ years of experience developing Python and C/C++ software close to hardware, plus master's research in RF systems, DSP and FPGA-oriented implementation. Strong fit for quantum control and calibration automation: experienced in turning experimental or device-level behaviour into repeatable software workflows, analysing measurement data, building validation tooling and collaborating across hardware/software boundaries. Background includes RF power-amplifier modelling, Digital Predistortion, sensor systems, telemetry, data acquisition and simulation-based verification.

Technical Skills

Python & Analysis: Python, NumPy-oriented modelling, data analysis, automation scripts, validation workflows, structured reporting, reproducible engineering notebooks/scripts

Measurement & Automation: data acquisition, sensor validation, telemetry pipelines, experimental workflow automation, reliability checks, performance metrics, hardware/software integration

RF, DSP & Control: RF power-amplifier behavioural modelling, Digital Predistortion, signal processing, nonlinear systems, calibration-oriented modelling, fixed-point analysis

Embedded & Systems: C/C++, Linux, shell scripting, embedded Linux, real-time embedded systems, communication interfaces, equipment monitoring, device integration

FPGA & Verification: VHDL, RTL simulation, pipelined datapaths, model-to-RTL validation, reference models, testbenches, Git, debugging, CI/CD-oriented workflows

Professional Experience

Embedded & Software Engineer — IoTaq Technology

Oct 2022 – Feb 2026

- Developed C/C++ and Python software for embedded and Linux-based telemetry systems used in equipment monitoring and diagnostics.
- Built data acquisition and communication pipelines between devices, sensors and backend systems, improving visibility into field equipment behaviour.
- Created Python tooling for data analysis, validation and reliability checks over telemetry outputs and equipment datasets.
- Debugged hardware/software integration issues, communication failures and performance problems across deployed embedded systems.
- Worked across device, Linux and backend layers to turn hardware behaviour into maintainable software workflows and operational metrics.

Engineering Intern — Lactec

2022

- Developed and tested embedded measurement systems and capacitive sensor prototypes, including data acquisition, validation and hardware/software integration.
- Supported experimental testing workflows, measurement repeatability checks and engineering analysis for sensor-based systems.

Research, Measurement & Automation Projects

RF Power-Amplifier Modelling and Digital Predistortion — Master's Research

- Developed Python reference models for RF power-amplifier behaviour and Digital Predistortion using memory polynomial structures.
- Analysed experimental datasets to extract model performance metrics such as NMSE and evaluate precision/complexity trade-offs.
- Built repeatable validation flows comparing Python models, fixed-point representations and VHDL simulation outputs.
- Worked with signal conditioning, nonlinear response modelling, coefficient estimation, calibration-like parameter selection and measurement-driven evaluation.
- Translated algorithmic results into hardware-aware implementation decisions, including pipelining, latency, quantisation and resource usage.

FPGA-based TRNG / ASIC Research

- Worked on a true random number generator described in VHDL, including RTL understanding, modification and validation in a microelectronics research context.

DSP and Wireless Systems Research — GICS / UFPR

- Conducted research in signal processing for wireless systems, RF optimisation and hardware-aware DSP implementation; presented research at Seminários de Microeletrônica do Paraná.

Baja SAE — Electrical and Embedded Systems

- Designed wiring harnesses and PCB-related subsystems; developed firmware and supported integration testing between electrical, mechanical and embedded subsystems.

Education

Master of Electronic Engineering — Universidade Federal do Paraná (UFPR), Brazil

Expected Aug 2026

Research focus: RF power amplifier behavioural modelling, Digital Predistortion, memory polynomial models, measurement-driven DSP evaluation and FPGA/VHDL implementation.

Bachelor of Electronic Engineering — Universidade Federal do Paraná (UFPR), Brazil

Relevant background: digital systems, microelectronics, embedded systems, signal processing, firmware, sensors, PCB design and electrical systems.